

OcuLayer: a zero-install browser viewer for segmented ocular micro-tomography

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Software

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Summary

OcuLayer is a browser application for exploring segmented ocular micro-computed-tomography (μ CT) data in two linked 3D views. The left pane holds a published, citable reference eye model; the right holds the toggleable segmented tissue layers of a scanned specimen. Orbiting one view mirrors onto the other, so a segmented coat can be read against the anatomy it matches, and every structure can be hidden, recoloured, faded or clipped with slice planes. It is a static site opened from a URL — no installation, no account, no gigabyte download. Its dataset is a CSV manifest supplied at run time, and a researcher's own meshes can be dropped onto the page and rendered beside the bundled data without leaving the browser.

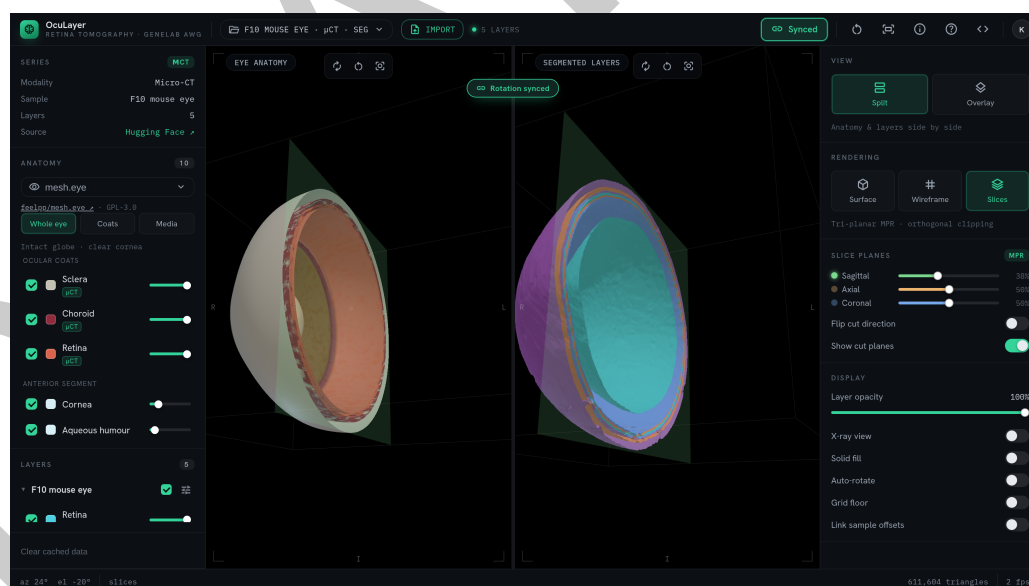


Figure 1: The two panes under a shared sagittal cut, cameras linked. Left: the mesh.eye reference model, ten structures each independently styled. Right: four of the five segmented μ CT coats of a murine eye, each separately streamed.

Statement of need

Segmented μ CT of the eye produces surface meshes far too large to open casually. The mouse eye shipped here is a 1.0 GB binary STL of 21.1 million triangles, and viewing it conventionally costs a desktop install, the full download and a machine able to hold it. That is reasonable for an investigator who works with the data daily, and prohibitive for the researchers, reviewers and

students who need to look at it once. This matters for space biology: spaceflight-associated neuro-ocular syndrome is among the better-documented risks of long-duration spaceflight (Lee et al., 2020), so ocular tissue recurs in spaceflight and analog studies. NASA's Open Science Data Repository publishes these data openly (Berrios et al., 2021; Gebre et al., 2025). But publishing a mesh is not the same as making it explorable: the repository hands back a file, and the barrier to looking at it is unchanged.

State of the field

3D Slicer (Fedorov et al., 2012) and ITK-SNAP (Yushkevich et al., 2006) do more than this viewer, including named segments, per-segment visibility, colour and opacity, and synchronised cameras. Both are desktop installs, and Slicer's user guide advises "10x more memory than the amount of data that you load" (3D Slicer Community, 2026). In the browser, NiiVue (Eckstein et al., 2026; Hanayik et al., 2026), Neuroglancer (Maitin-Shepard et al., 2021) and itk-vtk-viewer (McCormick et al., 2024) render meshes client-side, and all three already bind a dataset at run time as this viewer does: NiiVue's web app and itk-vtk-viewer take mesh URLs as ?images= and ?fileToLoad= respectively, and a Neuroglancer scene is a pasteable link. The Open Anatomy Browser (Halle et al., 2017) is closest, being zero-install, manifest-described, named and static. We claim novelty in none of this.

What none supplies is the other half of the comparison. None of the Open Anatomy atlases is ocular, and of six related open projects we surveyed (ISETBio (Cottaris et al., 2019), OpenRetina (D'Agostino et al., 2026), V-Cornea (Vanin et al., 2025; Vanin, 2025), OpenEyeSim (Priamikov et al., 2016), pulse2percept (Beyeler et al., 2017) and Open Source Brain (Gleeson et al., 2019)) none ships usable 3D eye geometry. This viewer therefore ships its own: three published eye models with per-structure names and provenance, in a second camera-linked pane. A multi-toggle layer panel would have been a fair contribution to itk-vtk-viewer, whose geometry panel selects one mesh at a time. The second pane would not: itk-vtk-viewer and the Open Anatomy Browser each build one model scene — one vtkProxyManager, and one model THREE.Scene beside an axes-only inset scene — so a second populated scene changes a central assumption instead of extending it. Neuroglancer and NiiVue already link cameras across panes, but neither reads glTF, the format of this viewer's bundled meshes. Hence a small library over three.js (Cabello & others, 2024), not a fork.

Software design

The site is bare ES modules behind an import map, and the deploy workflow uploads the repository tree itself, so a reviewer reads the files the browser runs. The price of refusing a bundled build is that three (r169) and the Draco decoder (1.5.7) are fetched from public CDNs and load-bearing for the deployed viewer.

The core library cannot tell that it is running in a browser. Renderer and controls enter through injected adapters, bytes through an injected io, state leaves only through an event emitter, and a single module constructs the WebGL context. Freedom from the DOM is enforced by a denylist scan over core/ rather than proved. In exchange the geometry, clipping and loading logic runs under Node in seconds.

Data reaches the viewer by one path whatever its source: the default manifest, a manifest URL typed into the Import dialog or passed as ?dataset=, and files dropped onto the page all become the same sample records. Dropped meshes are served to the core through an io wrapper that reads them from memory, so the core never learns that a File exists and nothing is uploaded. A manifest URL is written back into the address bar, which makes a view shareable as a link; a drop is not, because the files it names live on one machine.

Decimation answers a client cost, not a hosting one: the viewer does load a 1 GB, 21 M-triangle

STL, but reading and parsing it took about 12 s and 2.4 GB; the decimated copy loads in under a second. A documented pipeline built on glTF Transform (McCurdy, 2026) converts binary STL to glTF, welds it into an indexed mesh, decimates it with the meshoptimizer simplifier (Kapoulkine, 2024) and compresses it with Draco (Google, 2024); layers stream on toggle into the Cache Storage API, so each mesh downloads at most once per browser. tools/bench measures the cost as a symmetric point-to-surface distance, area-weighted over both meshes and normalised by the bounding-box diagonal:

Mesh	Triangles	Source	Shipped	Reduction	Mean error	p99	Area
eye	21.1 M	1008 MB	633 KB	1631×	0.017%	0.062%	+0.63%
feature	3.13 M	149 MB	325 KB	471×	0.006%	0.024%	+0.21%

Discarding 98.5% of eye.stl's triangles moves the surface by 0.017% of the diagonal on average; the worst-case (Hausdorff) distances, 3.99% and 1.10%, fall almost entirely in the original-to-decimated direction, consistent with fragments removed rather than the principal surface displaced. The result is an instrument for orientation and triage, not morphometry. A first visit transfers a 213 KB shell, served gzipped at 69 KB; loading the default reference eye adds 343 KB, against 1.13 GB of source meshes, with three, the webfonts and the Draco decoder fetched from CDNs on top.

The left pane holds third-party published anatomy, not NASA data: feelpp/mesh.eye (Chabannes et al., 2025; Sala et al., 2024), the SolidWorks CAD eye it derives from, and the Upatras OpenSim oculomotor model with its six extraocular muscles (Filip et al., 2018), each with one named node per structure under its own upstream licence. Because these models are human while the segmentation is murine, the interface states that the left pane is for orientation, not cross-species morphometry.

Research impact statement

The NASA GeneLab Analysis Working Group, for whom the viewer was built, has used it to load the segmented μ CT data and compare structures across it; the specimen in Figure 1 is that data.

The viewer is deployed at https://gojian.github.io/awg_retina_tomography_ui/, and the data behind it are open and ungated. The segmented meshes, the CSV manifest, the source reconstruction slices and the full-resolution meshes the shipped assets were decimated from are published under MIT on Hugging Face (Shah, 2025), so a reader can recompute the reduction factors and error bounds above instead of taking them on trust. tools/bench uses a fixed seed, so runs are reproducible.

optimized/anatomy/README.md is a licence-and-geometry audit of nine entries from eight open projects: per-structure triangle counts, volumes and upstream DOIs for the three that distribute usable 3D eye geometry (Chabannes et al., 2025; Filip et al., 2018; Sala et al., 2024), and a recorded check for the six that do not. That geometry is other groups' published work, carried under its own GPL-3.0 and CC BY 4.0 terms; MIT covers the viewer code only.

552 unit tests on Node's built-in runner cover the data and caching layers, the file import, the geometry code behind the error figures above, and the core library, run headless under a stub renderer on synthetic STL and uncompressed glTF. 22 Playwright tests drive the real application in Chromium, including a file import from disk and a manifest loaded by URL. CI runs both suites and decodes every shipped asset, so a corrupt mesh fails the build.

AI usage disclosure

Generative AI was used in developing this software and preparing this submission: Claude (Anthropic), from November 2025 to October 2026, through Claude Code from June 2026, using Claude Opus 4.8 (claude-opus-4-8), Claude Opus 5 (claude-opus-5), Claude Fable 5.1 (claude-fable-5-1) and Claude Opus 5.5 (claude-opus-5-5).

- *The application.* The first viewer (November 2025 to January 2026) was written with Claude (model version not recorded). From June 2026 it was rewritten with AI assistance: Claude Opus 4.8 wrote `viewer.js`, `asset-loader.js`, the decimation pipeline and the shipped μ CT meshes; Opus 5 the anatomy-model build scripts; Opus 5.5 the file import.
- *Core-library refactor.* The DOM-free core/ library was extracted from `viewer.js` with Claude Fable 5.1, following a written architecture plan the authors reviewed; each step was gated on the full test suite and verified against the unchanged end-to-end tests.
- *Tests.* The unit and Playwright suites were generated with AI assistance from the authors' description of the intended behaviour.
- *Benchmark tooling.* `tools/bench/` was implemented with AI assistance. Its correctness rests on the geometry tests, which cross-check the spatial index against brute force, and the numbers reported here are the tool's output, not the model's.
- *Eye-model survey.* The survey of open eye-modelling projects and the licence-and-geometry audit in `optimized/anatomy/README.md` were done with Claude Opus 5.
- *Documentation and paper.* The README, CONTRIBUTING.md, docs/api.md, the changelog, the citation file and this paper were drafted with AI assistance and edited by the authors; the figure was captured by an AI session driving the application. Bibliographic entries were checked against Crossref, arXiv and Zenodo with AI assistance and confirmed by the authors.

The authors ran the tests and benchmarks through Claude Code and checked the results, reviewed, edited and validated all AI-assisted output, and made the core design decisions: the two-pane linked-view concept, shipping decimated Draco assets with a documented accuracy budget, using published open eye models as reference anatomy, and the in-browser import of a user's own data. The authors take full responsibility for the accuracy, originality and licensing of all submitted material.

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